

Introduction to Computer and Communication Network

Project Deliverable 3

TASK 1:

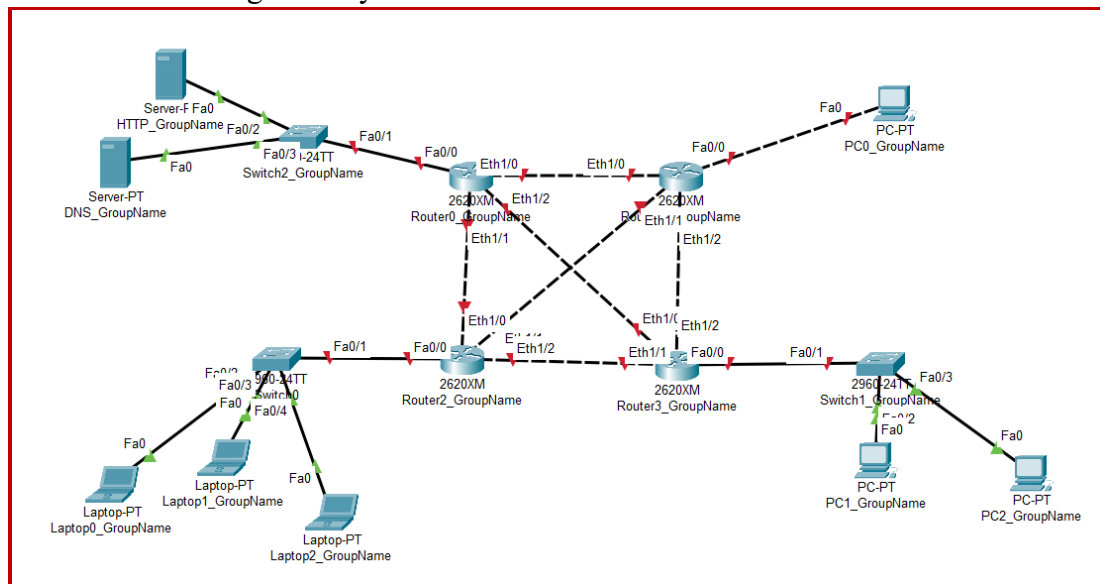
Start a project in Cisco Packet Tracer and answer the following questions:

[50 points]

(NOTE: place your screenshots in each of your answers).

(NOTE: replace “GroupNumber” below with the number of your group. You will lose points for each question in which the screenshot does not show your group number.)

Build the network given to you below:



Apply all the required settings on the following network's devices using **static routing** between the routers in order **all devices** are able to communicate to each other. Assign your group domain name to your HTTP server. Show all the steps you made and at the end test connectivity and access HTTP server from PC2 using web browser.

Step 1: Network Setup

Build the network topology as shown in the figure provided.

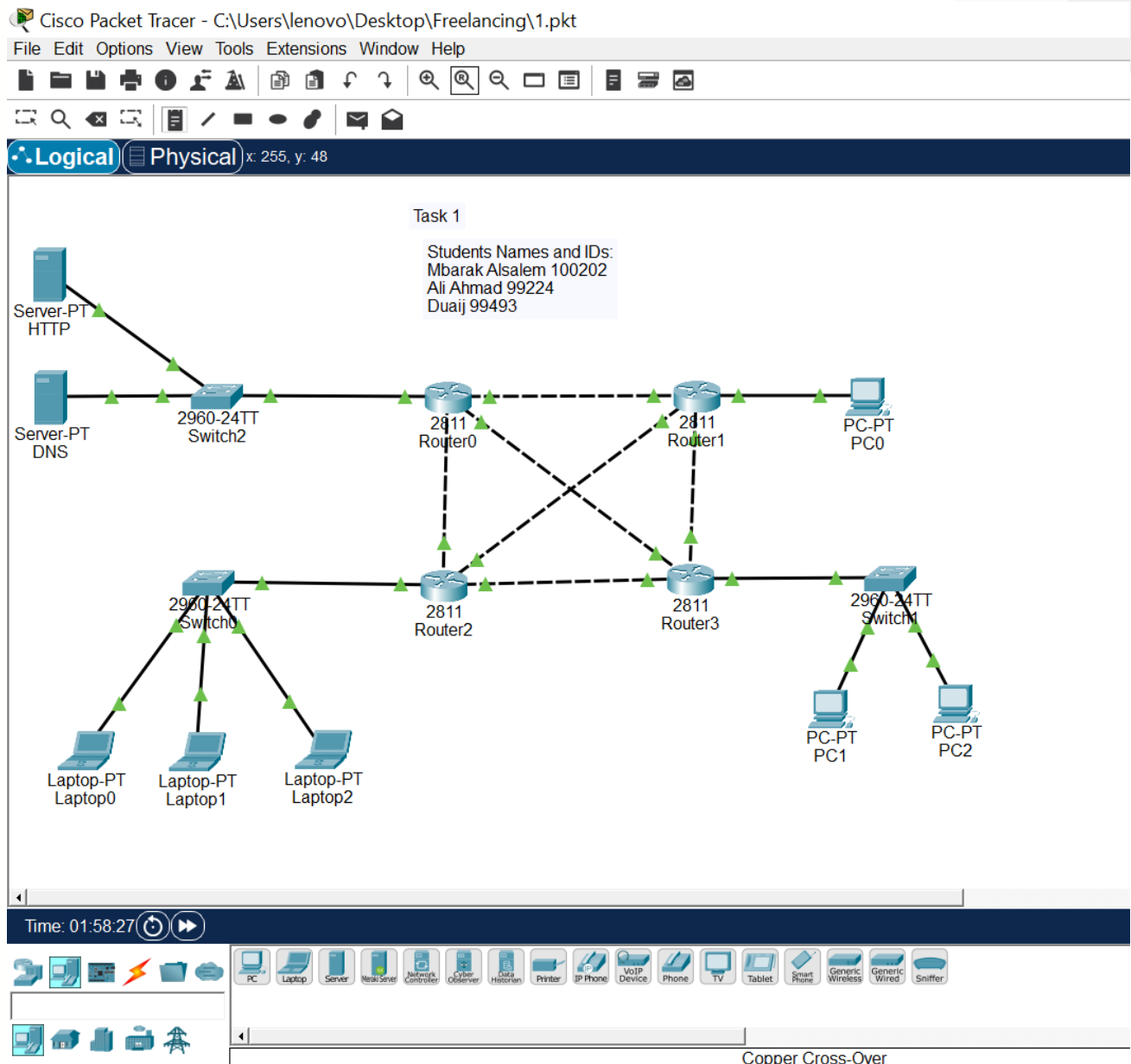


Figure 1: Full topology

Step 2: IP Addressing

1. What is the total number of **networks/subnets** in the topology?

Answer:

Number of subnets:

9 subnets (or count based on your actual topology)

Network addresses table:

Subnet	Network Address	Usable Range
1 (Router0-Router1)	192.168.1.0/24	192.168.1.1 - 254
2 (Router0-Router3)	192.168.2.0/24	192.168.2.1 - 254
3 (Router1-Router2)	192.168.3.0/24	192.168.3.1 - 254

4 (Router2-Router3)	192.168.4.0/24	192.168.4.1 - 254
5 (Switch2 - Servers)	192.168.10.0/24	192.168.10.1 - 254
6 (Switch0 - Laptops)	192.168.20.0/24	192.168.20.1 - 254
7 (Switch1 - PC1,PC2)	192.168.30.0/24	192.168.30.1 - 254
8 (Router1-PC0)	192.168.40.0/24	192.168.40.1 - 254

2. For **each** network/subnet:

List the **network address you will use**:

Example: Subnet 1 → 192.168.1.0/24 → Usable range: 192.168.1.1 – 192.168.1.254

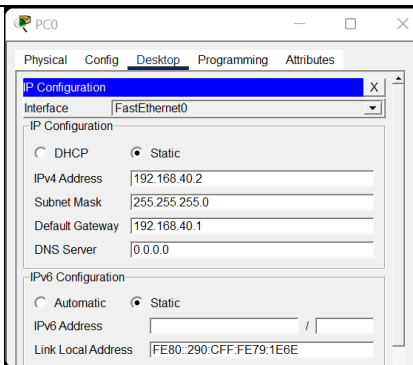
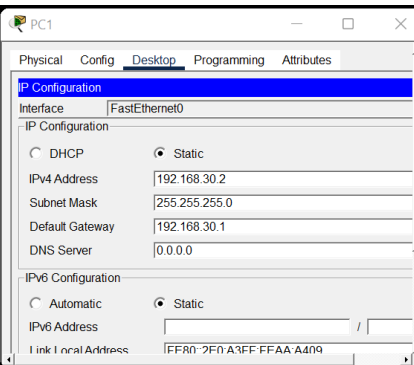
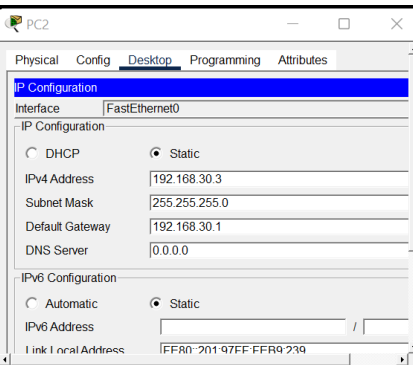
Step 3: Device Configuration

Configure all network devices:

- PCs / Laptops
- Routers
- Servers

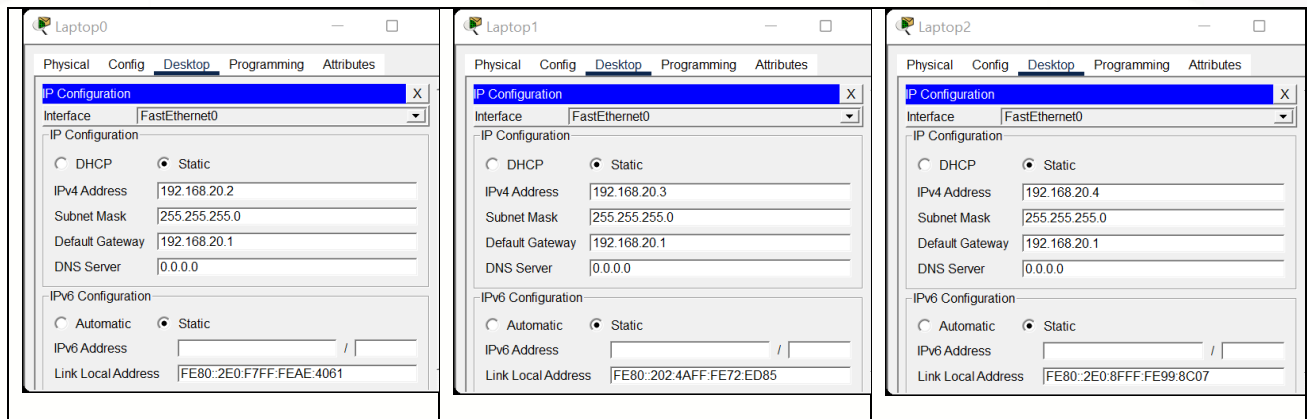
Device	What to screenshot
Router0	Config tab → each interface (Fa0/0, Fa0/1, Fa1/0)
Router1	Config tab → each interface (Fa0/0, Fa0/1, Fa1/0)
Router2	Config tab → each interface
Router3	Config tab → each interface

PC IP Config:

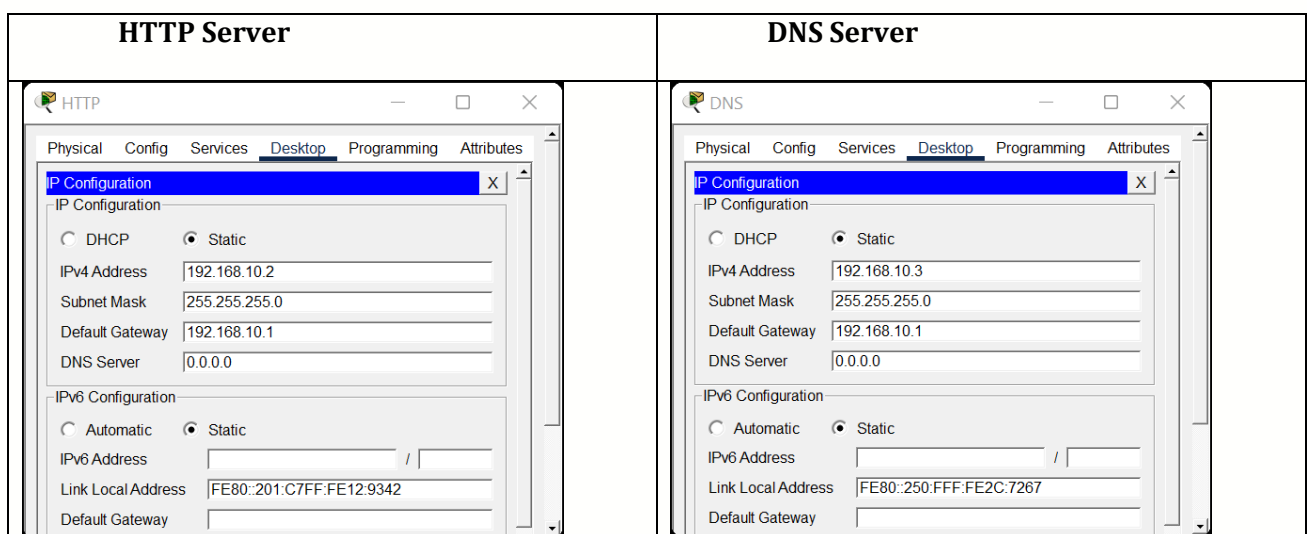
PC0	PC1	PC2
		

Laptop Ip config:

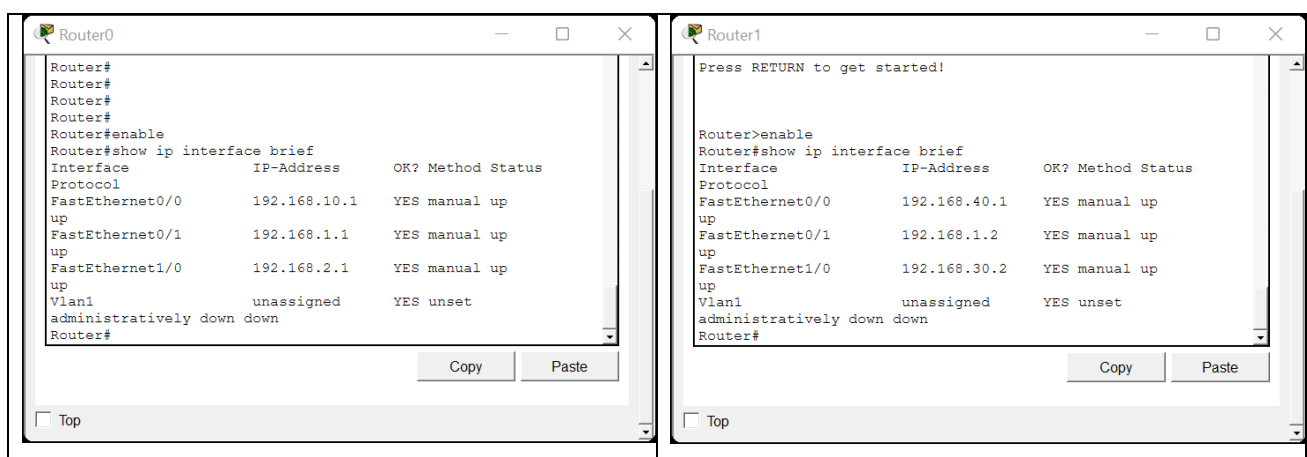
Laptop0	Laptop1	Laptop2
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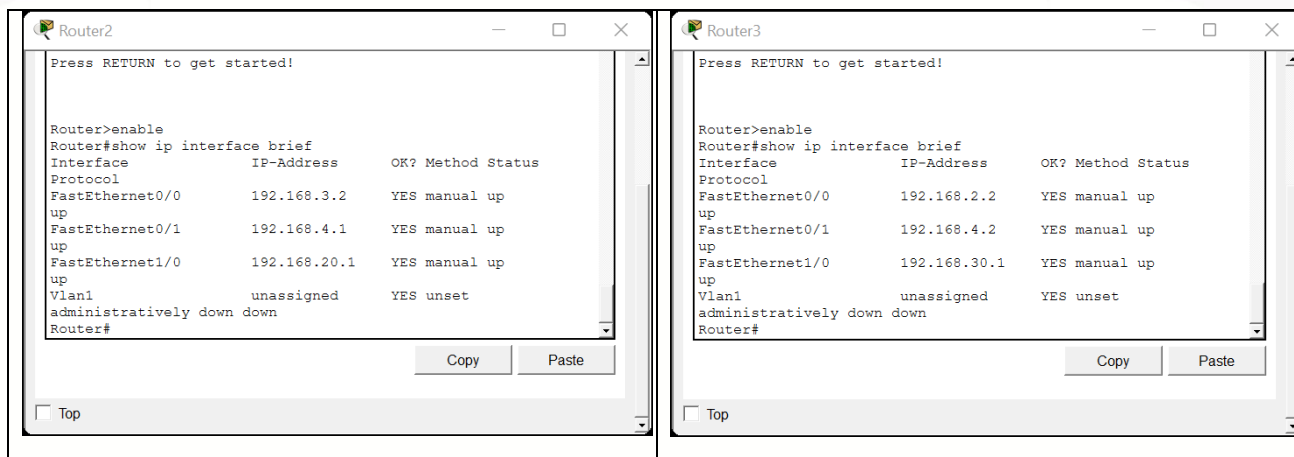


Servers IP Configurations:



Routers IP Config



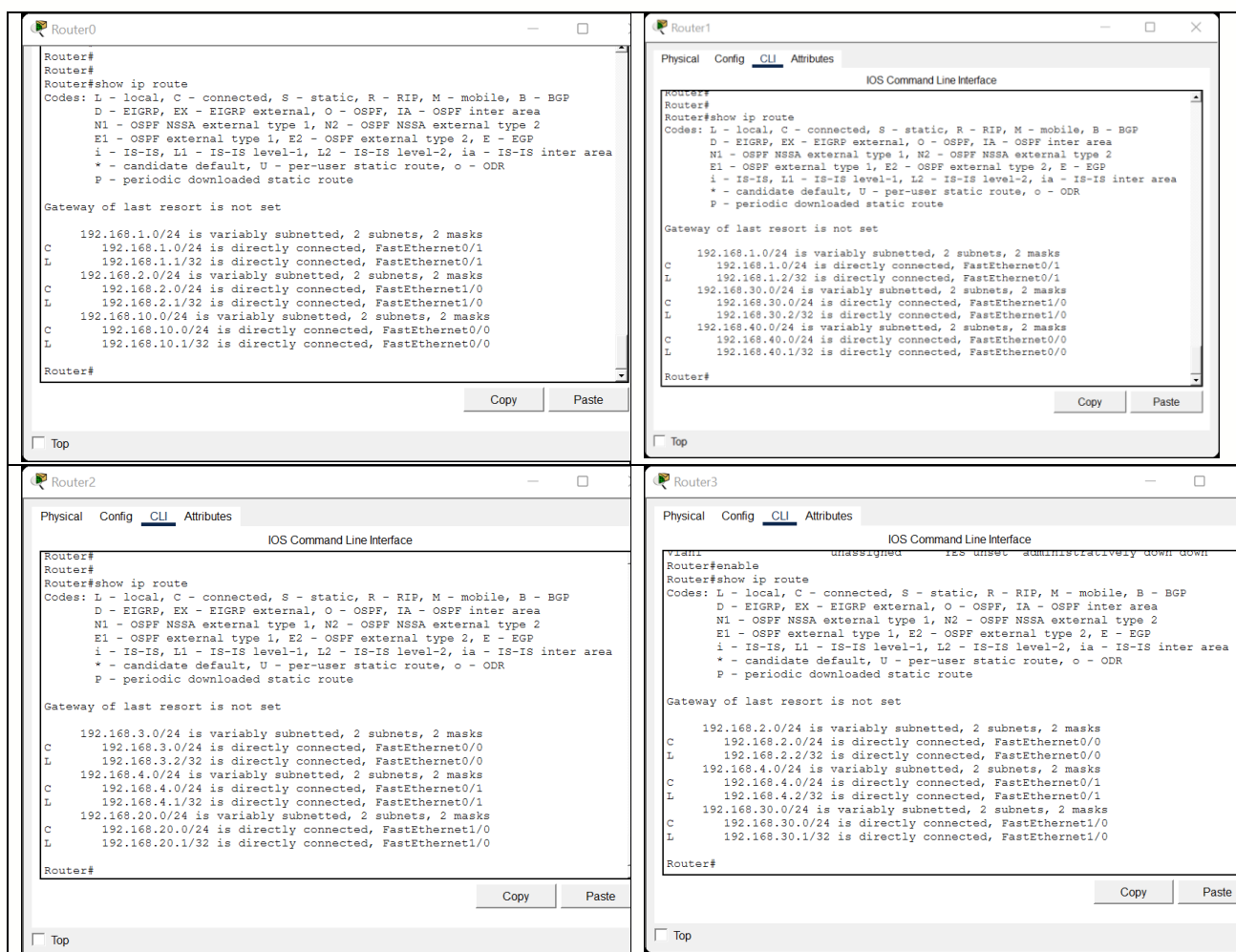


Step 4: Static Routing

Configure static routes on each router so that all networks can communicate.

Include screenshots of routing configurations on each router.

show ip route

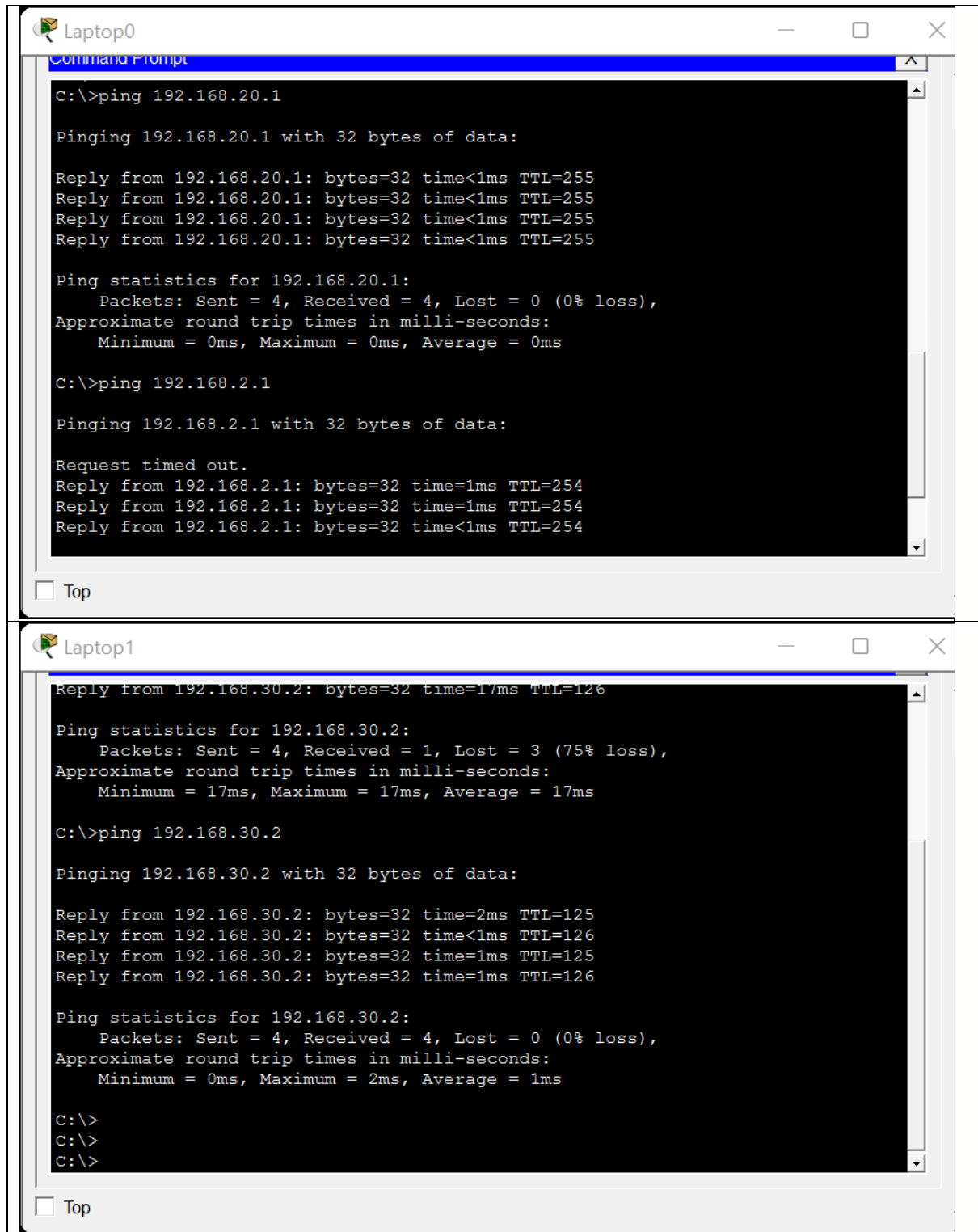


Step 5: Connectivity Testing

Test connectivity using the command line:

- Laptop0 → PC0
- Laptop1 → PC1

Include screenshots



Step 6: Web Server Configuration

- Assign the web server (HTTP) a **domain name** using your group number (Example: group1.com)
- From **PC2**, access the web server using a browser

Include screenshot showing successful access.

